Optimization-based precipitation estimation from commercial microwave links

Thesis defense

Iván Soto
Saarland University

Glossary: Microwave Links

Commercial microwave link (CML).

A fixed wireless connection between two antennas, used for communications backhaul. In this talk, a link is represented by a line segment ℓ.

Frequency.

The carrier frequency of the microwave signal, measured in GHz.

Polarization.

The orientation of the electric field of the microwave signal.

Glossary: Rain Field and Attenuation

Rain field.

A function from a geographic area Ω to nonnegative rain rates:

$$R : \Omega \longrightarrow \mathbb{R}_{\geq 0}.$$

For a pixel $s \in \Omega$, $R(s)$ is the rain rate, measured in mm/h.

Rain-induced attenuation.

Raindrops absorb and scatter microwave energy, weakening the received signal. Rain-induced attenuation is the additional signal loss along a link, measured in dB.

Specific attenuation. Signal loss per unit distance, measured in dB/km.

Glossary: Forward Model

Forward model.

A known function that maps the hidden object to predicted measurements:

$$\text{object} \longrightarrow \text{measurements}.$$

In our case.

The hidden object is the rain field R. The measurements are the link attenuations:

$$R \longrightarrow \{A_\ell\}_\ell.$$

Glossary: Inverse Problem

Inverse problem.

Estimate the hidden object from its measurements:

$$\text{measurements} \longrightarrow \text{estimated object.}$$

In our case.

Given link attenuations and link properties, estimate the rain field:

$$\{A_\ell\}_\ell \longrightarrow \hat{R}.$$

Ground truth and reconstruction.

The benchmark reference field is R. An algorithm returns a reconstruction $\hat{R}$.

Glossary: Evaluation Metrics

Let R_i and $\hat{R}_i$ be the reference and reconstructed rain rates at pixel i, for n paired pixels.

$$\mathrm{Bias} = \frac{1}{n}\sum_{i=1}^{n}(\hat{R}_i - R_i) \qquad \text{mean signed error}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{R}_i - R_i)^2} \qquad \text{root mean squared error}$$

Both metrics have units of mm/h. Positive bias means overestimation.

Pearson correlation. Spatial linear association between the centered maps:

$$\rho = \frac{\sum_i (\hat{R}_i - \bar{\hat{R}})(R_i - \bar{R})}{\sqrt{\sum_i (\hat{R}_i - \bar{\hat{R}})^2}\sqrt{\sum_i (R_i - \bar{R})^2}}.$$

How much is it raining outside?



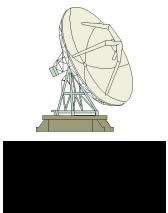
Rain gauges

Collect precipitation at a location and measure its accumulated depth.

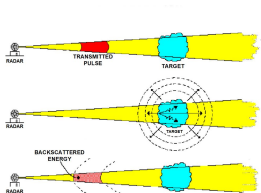
Direct point measurements, but sparse coverage between gauges.

Weather radar

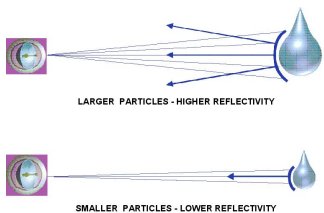
Scanning antenna



Transmitted pulse and echo

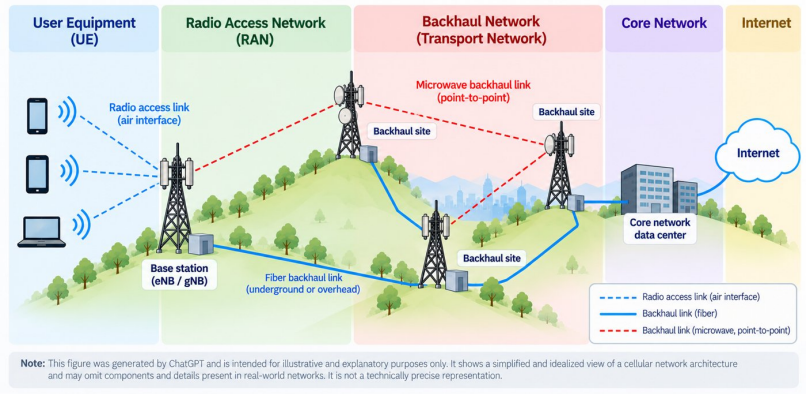


Liquid-target reflectivity



- Reflectivity is converted into an estimated rain rate.
- Broad spatial coverage, but the reflectivity-to-rain-rate conversion introduces uncertainty.

Commercial microwave links in a cellular network



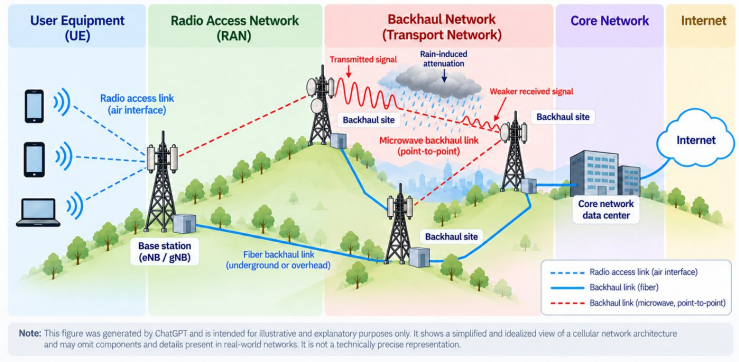
Red dashed connections: point-to-point microwave backhaul.
They carry users' communications and network signaling.

Milestones in microwave rainfall sensing

1977–1978	**Atlas & Ulbrich; Olsen et al.** Physical basis: near-linearity at selected frequencies and power-law models.
1991	**Giuli et al.** Tomographic rainfall-field reconstruction from simulated link attenuations.
2000–2003	**Holt et al.; Rahimi et al.** Dedicated dual-frequency links: path rainfall compared with gauges and radar.
2006–2007	**Messer et al.; Leijnse et al.** Rainfall retrieval from commercial links in Israel and the Netherlands.
2013 / 2016	**Overeem et al.** Netherlands-wide maps using ordinary kriging; later, 2.5 years of maps.
2020	**Graf et al.** German-wide evaluation with approximately 4,000 CMLs and IDW.

Our comparison uses IDW, an established baseline in large-scale CML rainfall mapping.

Rain-induced attenuation



- When it rains, the microwave signal loses power.
- We use this signal loss to estimate rainfall along the link.

The rain-attenuation forward model

Specific attenuation

$$\gamma_\ell(R) = \kappa_\ell R^{\alpha_\ell} \qquad [\mathrm{dB/km}].$$

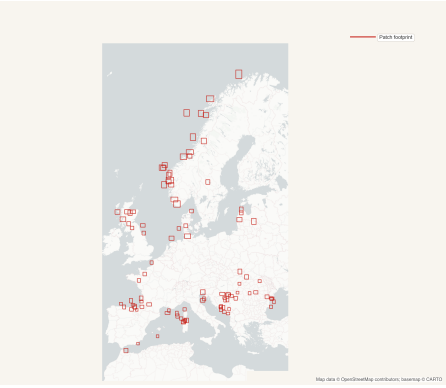
Attenuation along a link

$$A_\ell(R) = \int_\ell \kappa_\ell R(s)^{\alpha_\ell}\, ds = \sum_{p \in \mathcal{C}(\ell)} d_{\ell,p} \kappa_\ell R_p^{\alpha_\ell}.$$

$d_{\ell,p}$: crossed length in pixel p, in km.

R_p: rain rate in mm/h.
Frequency and polarization determine κ_ℓ and α_ℓ.

Benchmark construction



100 precipitation patches

84 hourly EURADCLIM maps
18–31 January 2023
Native grid: 2×2 km

4TU Netherlands network template
939–1,996 observations per patch

Reference fields and synthetic measurements

Refinement and smoothing

Each 2-km cell becomes 16×16 inherited subpixels, at 125-m spacing.

$$R^{\star} = K_{\sigma} * R^{\uparrow}, \qquad K_{\sigma}(i,j) = \frac{\exp[-(i^2+j^2)/(2\sigma^2)]}{\sum_{a,b=-3}^{3} \exp[-(a^2+b^2)/(2\sigma^2)]}, \quad \sigma = 1.$$

Synthetic attenuation

Overlay the network template and apply the rain-attenuation model to $R^{\star}$.

Statistic	Value (mm/h)
Mean of the 100 spatial patch means	To be computed

Reconstruction methods

Method	Main idea
IDW	Treat each link midpoint as a virtual rain gauge.
ILDW	Weight observations by distance to the link segment.
Regularized optimization	Estimate nonnegative pixel rain rates using attenuation agreement, smoothness, and shrinkage.

$$\min_{r \geq 0} \ \alpha_{\text{atten}} J_{\text{atten}}(r) + \alpha_{1d} J_{1d}(r) + \alpha_{2d} J_{2d}(r) + \alpha_{\text{total}} J_{\text{total}}(r).$$

A counterexample to convexity

One horizontal link crosses the middle row of a 3×3 grid.

$$R^{(L)} = \begin{bmatrix} 1 & 1 & 1 \\ 20 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \qquad R^{(R)} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 20 \\ 1 & 1 & 1 \end{bmatrix}, \qquad R^{(M)} = \frac{R^{(L)} + R^{(R)}}{2}.$$

Both endpoint fields produce $A_{\mathrm{obs}} = 0.80155$ dB.

$$J_{\mathrm{atten}}(R) = \frac{(\widehat{A}(R) - A_{\mathrm{obs}})^2}{|\ell|}, \qquad J_{\mathrm{atten}}(R^{(L)}) = J_{\mathrm{atten}}(R^{(R)}) = 0.$$

For $0 < \alpha < 1$, strict concavity gives $2(10.5)^{\alpha} > 20^{\alpha} + 1$.

$$J_{\mathrm{atten}}(R^{(M)}) = 0.0050431 > \frac{J_{\mathrm{atten}}(R^{(L)}) + J_{\mathrm{atten}}(R^{(R)})}{2} = 0.$$

Convex surrogate using virtual links

Replace each link by a virtual observation with the same geometry and a linear attenuation model.

$$R_\ell^{\mathrm{eq}} = \left(\frac{A_\ell}{\kappa_\ell |\ell|} \right)^{1/\alpha_\ell}, \qquad A_\ell^0 = \kappa_0 R_\ell^{\mathrm{eq}} |\ell|.$$

$$\widehat{A}_\ell^0(r) = \kappa_0 \sum_p d_{\ell,p} r_p.$$

The squared residuals are convex in r. Positive squared ℓ_2 shrinkage makes the surrogate strictly convex. Homotopy continuation gradually returns to the original link frequencies.

Whole-patch reconstruction results

Method	RMSE (mm/h)	Signed bias (mm/h)	Pearson r
IDW	1.307 ± 1.234	0.034 ± 0.296	0.866 ± 0.059
ILDW	1.239 ± 1.182	0.047 ± 0.261	0.878 ± 0.055
Solver(ILDW)	1.059 ± 1.192	0.040 ± 0.235	0.918 ± 0.055
Convex Solver	0.845 ± 0.803	0.023 ± 0.075	0.940 ± 0.034
Homotopy Solver	0.845 ± 0.803	0.023 ± 0.075	0.940 ± 0.034
Solver(GT)	0.503 ± 0.369	0.009 ± 0.030	0.961 ± 0.032

Mean $\pm$ standard deviation across 100 patches.

Solver(GT) is an optimistic diagnostic, initialized from the reference field.

Conclusions and outlook

Controlled benchmark
Optimization improves mean RMSE and spatial correlation relative to IDW. **Operational validation**
Test measured attenuation, baseline estimation, wet-antenna effects, and measurement noise. **Temporal reconstruction and nowcasting**

Use optical flow to forecast a reconstructed field, then revise it when new measurements arrive. **Learned rainfall priors**

Explore Bayesian reconstruction with a learned prior:

$$p(R \mid A) \propto p(A \mid R)\, p(R).$$